

Diagnosing Estimator Homomorphism Compatibility in Reinforcement Learning with Verifiable Rewards

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Abstract

Reinforcement learning with verifiable rewards (RLVR) evaluates decoded outcomes, whereas policy-gradient estimators operate on concrete trajectories and batch-dependent training state. These two levels need not respect the same equivalence relation. We formalize *estimator homomorphism compatibility*: when several concrete trajectories represent the same reward- or environment-level outcome, the stochastic estimator should not introduce an expected update component that distinguishes representatives solely through estimator state. For exchangeable score estimators with batch-coupled scalar coefficients, we derive an exact quotient/fiber decomposition in which the within-class component is governed by the covariance between a representative-conditioned effective coefficient and the conditional fiber score. Exact enumeration validates the identity. Controlled Transformer experiments, a frozen Hugging Face TRL `GRPOTrainer`, whole-model training, and cross-family tests show measurable estimator-induced representation pressure. An equal-length control demonstrates that the measured compatibility gap is not reducible to response-length weighting: two one-token strings accepted as the same answer can enter different clipping states after policy lag and yield different optimizer updates. Lower-learning-rate whole-model and LoRA follow-ups reproduce selective clipping in additional settings. We do not show that the effect necessarily harms downstream reward. The contribution is a quotient-conditioned estimator diagnostic, a decomposition that organizes multiple known estimator mechanisms, and direct measurements in LLM RLVR training.

1 Introduction

RLVR has become a practical route to improving reasoning in large language models, with Group Relative Policy Optimization (GRPO) and related estimators widely used in post-training [Shao et al., 2024, Liu et al., 2025]. Outcome-based rewards, however, evaluate an abstract result: a numerical answer, a verifier decision, or another decoded property. The optimizer acts on a concrete au-

toressive trajectory together with its tokenization, normalization denominator, likelihood ratios, clipping state, and other batch-level quantities.

This separation raises a simple question: *if two trajectories are equivalent to the reward or environment, does the stochastic training estimator treat them as equivalent?* Existing work already establishes several mechanisms that can make policy-gradient contributions representation dependent, including response-length normalization, sampled token denominators, sequence-level importance ratios, and PPO-style clipping [Schulman et al., 2017, Liu et al., 2025, He et al., 2025, Zeng et al., 2026, Ding et al., 2026, Zheng et al., 2025b]. The goal of this paper is not to claim those mechanisms as new. Instead, we ask whether they can be described and measured through one quotient-conditioned compatibility object.

Let $Q(Y)$ map a concrete trajectory Y to the outcome distinguished by a reward or environment. We study whether a stochastic estimator used by a concrete trainer *descends to the same quotient*. The central object is the representative-conditioned effective coefficient

$$h_\theta(y) = \mathbb{E}[H_i \mid Y_i = y], \quad (1)$$

where H_i is the scalar coefficient multiplying the score of sample i after accounting for reward, advantage, normalization, importance weighting, clipping, and batch state.

Our contributions are:

1. an exact quotient/fiber decomposition for exchangeable score-function estimators with batch-coupled scalar coefficients;
2. a compatibility gap $\Gamma_\theta(z)$ that measures expected within-equivalence-class update and a class-constancy criterion for eliminating that gap;
3. direct measurements on tokenization fibers and verifier-defined trajectory classes in a frozen production-style `GRPOTrainer`; and
4. an equal-length clipping control showing that the measured compatibility failure is not specific to response-length weighting, together with lower-learning-rate whole-model and LoRA robustness checks.

We use *Tokenization-Fiber Gradient Leakage* (TFGL) for the special case in which multiple token trajectories

lie in the same decoded-reward class and the estimator induces a within-class update. TFGL is a diagnostic instance of the broader framework, not a claim that non-canonical tokenization is new.

2 Related Work

RLVR aggregation and length. GRPO was introduced in the DeepSeekMath line [Shao et al., 2024]. Dr.GRPO identifies response-length and question-difficulty biases in GRPO-style training [Liu et al., 2025]. Subsequent work analyzes aggregation and normalization more directly: ΔL Normalization studies minimum-variance unbiased aggregation [He et al., 2025], Balanced Aggregation analyzes token- versus sequence-level aggregation bias [Zeng et al., 2026], and Ding et al. [2026] proves that a broad family of length-based weighting rules cannot simultaneously satisfy gradient unbiasedness and length invariance. These results occupy broad length-bias novelty. We use length-dependent tokenization fibers as controlled diagnostics rather than as the paper’s novelty claim.

Importance ratios and clipping. PPO clips per-sample policy ratios [Schulman et al., 2017], while GSPO moves LLM RL toward sequence-level likelihood ratios and sequence-level clipping [Zheng et al., 2025b]. Most directly relevant to our equal-length control, Hakoda et al. [2026] show that independent PPO ratios/clipping can produce symmetry-inconsistent updates in a symmetric MDP and propose a coupled symmetric ratio. Therefore this paper does not claim that clipping-induced symmetry breaking is new. Our clipping experiment instead serves as a controlled falsification of the narrower hypothesis that the measured LLM compatibility gap is only a response-length effect.

Abstraction and homomorphism. MDP homomorphisms and homomorphic policy-gradient theory formalize how ground states/actions relate to abstract control problems [Rezaei-Shoshtari et al., 2022, Panangaden et al., 2024]. We hold the reward/environment quotient fixed and diagnose whether a finite-sample estimator implemented by a concrete trainer preserves it. The distinction is between defining an abstract objective and measuring symmetry breaking introduced by the estimator used to optimize it.

Tokenization and trajectory representation. A single string can have many non-canonical tokenizations, and modern language models can retain substantial competence under such segmentations [Zheng et al., 2025a]. We do not claim tokenization multiplicity as new. Instead, exact same-surface paths provide a convenient reward-null direction on which estimator dependence can be measured.

3 Problem Setup

Let $\mathcal{Y}$ be a finite or countable ground trajectory/action space and let $Q : \mathcal{Y} \rightarrow \mathcal{Z}$ be a quotient map. Write

$$C(z) = \{y : Q(y) = z\}, \quad P_\theta(z) = \sum_{y \in C(z)} \pi_\theta(y), \quad (2)$$

and, for $y \in C(z)$,

$$\rho_\theta(y | z) = \frac{\pi_\theta(y)}{P_\theta(z)}. \quad (3)$$

The ground score decomposes as

$$s_\theta(y) = \nabla_\theta \log \pi_\theta(y) = s_Z(z) + s_F(y | z), \quad (4)$$

where

$$s_Z(z) = \nabla_\theta \log P_\theta(z), \quad s_F(y | z) = \nabla_\theta \log \rho_\theta(y | z). \quad (5)$$

The conditional score identity yields $\mathbb{E}[s_F(Y | z) | z] = 0$.

Draw an iid batch $Y_1, \dots, Y_B \sim \pi_\theta$ and consider any exchangeable estimator of the form

$$\hat{U} = \sum_{i=1}^B H_i(Y_{1:B}) s_\theta(Y_i), \quad (6)$$

where each H_i is scalar and may depend on rewards, lengths, group baselines, random denominators, clipping state, policy ratios, and every other sample in the batch. We assume the relevant expectations are finite. Define

$$h_\theta(y) = \mathbb{E}[H_i | Y_i = y], \quad (7)$$

$$\bar{h}_\theta(z) = \mathbb{E}[h_\theta(Y) | Q(Y) = z]. \quad (8)$$

Exchangeability makes the definition independent of index i .

4 Estimator Homomorphism Compatibility

Theorem 1 (Exact quotient/fiber decomposition). *At a fixed parameter value θ ,*

$$\mathbb{E}[\hat{U}] = B \sum_{z \in \mathcal{Z}} P_\theta(z) [\bar{h}_\theta(z) s_Z(z) + \Gamma_\theta(z)], \quad (9)$$

where

$$\Gamma_\theta(z) = \mathbb{E}[h_\theta(Y) s_F(Y | z) | Q(Y) = z] \quad (10)$$

$$= \text{Cov}(h_\theta(Y), s_F(Y | z) | Q(Y) = z). \quad (11)$$

Proof. By exchangeability, $\mathbb{E}[\hat{U}] = B \mathbb{E}[H_1 s(Y_1)]$. Iterated expectation conditional on Y_1 gives $B \mathbb{E}[h(Y_1) s(Y_1)]$. Conditioning next on $Z = Q(Y_1)$ and substituting $s(Y) = s_Z(z) + s_F(Y | z)$ yields Eq. (9). Because $\mathbb{E}[s_F | z] = 0$, the second term is the conditional covariance in Eq. (11). $\square$

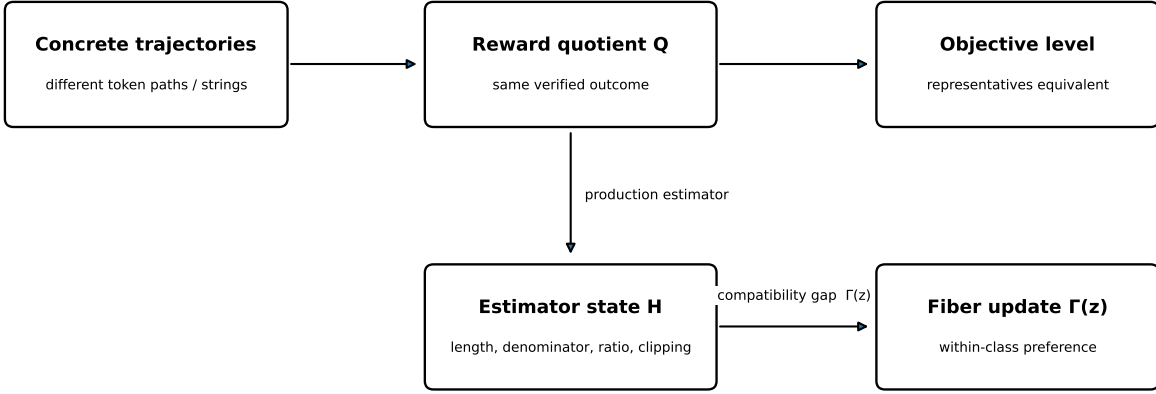


Figure 1: Objective-level equivalence need not imply estimator-level equivalence. A quotient Q identifies concrete trajectories as the same outcome, while a production estimator may attach representative-dependent effective coefficients through length, random denominators, policy ratios, clipping, or other state. The resulting within-class term is the estimator homomorphism gap $\Gamma(z)$.

Definition 1 (Estimator homomorphism compatibility). *The estimator is homomorphism-compatible at θ if $\Gamma_\theta(z) = 0$ for every quotient class z .*

This is an estimator property, not an objective property. An estimator may be compatible while still having quotient-level bias, an incorrect global scale, or high variance.

Corollary 1 (Class-constant coefficient). *If, for every class z , $h_\theta(y) = c_\theta(z)$ almost surely for $y \in C(z)$, then $\Gamma_\theta(z) = 0$.*

Class-constancy is sufficient for any parameterization. It is locally necessary only when the within-class score tangent space is rich enough to represent the centered coefficient $h_\theta(y) - \bar{h}_\theta(z)$; Appendix A states this condition precisely.

Constant realized sums are insufficient. A realized identity such as $\sum_i W_i = c$ only implies the marginal mean $\mathbb{E}[W_i] = c/B$. It does not imply $\mathbb{E}[W_i | Y_i = y] = c/B$ or class-constancy. For example,

$$W_i = \frac{a(Y_i)}{\sum_j a(Y_j)} \quad (12)$$

has deterministic realized sum one while its representative-conditioned expectation can depend on y .

Table 1 illustrates how known RLVR mechanisms enter the same diagnostic object. The point is not that these mechanisms are new, but that quotient compatibility asks the same conditional question of each one.

Mechanism	Representative-dependent state	Compatibility risk
GRPO response mean	response length	unequal-length representatives
BNPO/DAPO-style mean	sampled active-token denominator	finite-batch coupling
Dr.GRPO denominator	fixed max length	denominator itself class-constant
Sequence clipping	policy ratio / clipping state	may differ at equal length
Token clipping	token-local ratio state	token-indexed extension
Reward scaling	batch reward statistics	group-composition dependence

Table 1: Examples of mechanisms entering $h(y)$.

5 Experimental Design

Models. We use GPT-2-family checkpoints [Radford et al., 2019] for controlled diagnostics and DistilGPT2 for the majority of actual-trainer experiments. Cross-family checks use Qwen2.5-0.5B [Yang et al., 2024]. The Qwen ecological pilot uses GSM8K [Cobbe et al., 2021]. LoRA transfer uses the low-rank adaptation method of Hu et al. [2021].

Trainer implementation. Actual-trainer experiments use a frozen vendored Hugging Face TRL **GRPOTrainer** implementation [Hugging Face, 2026b]. Dropout is disabled for paired deterministic comparisons. The frozen commit and file hash are given in Appendix E.

Equivalence classes. We study exact same-decoded-string classes and verifier-defined classes. The former provide tokenization fibers; the latter include distinct strings such as "4" and "+4" that a numeric verifier maps

to the same outcome.

Protocol discipline. Target protocols are described as *prospectively frozen*: candidates, thresholds, seeds, or phase-diagram grids were fixed before observing the corresponding target result. We do not call them externally preregistered. Failed and implementation-invalid experiments remain in the record (Appendix C).

Dimensionless reporting. Raw parameter-space dot products are not comparable across subspaces. We therefore emphasize

$$R_{\perp} = \frac{\|\delta_{\perp}\|}{\|u_{\text{ref}}\|}, \quad (13)$$

$$P_F = \frac{\langle \delta, v_{\perp} \rangle}{\|v_{\perp}\| \|u_{\text{ref}}\|}, \quad (14)$$

$$C_F = \frac{\langle \delta_{\perp}, v_{\perp} \rangle}{\|\delta_{\perp}\| \|v_{\perp}\|}. \quad (15)$$

Bootstrap intervals below resample independent seeds and are descriptive, not substitutes for larger task-level replication.

6 Results

6.1 Tokenization fibers and numerical controls

Natural same-surface alternative paths occur with measurable probability in both DistilGPT2 and Qwen2.5-0.5B. On a prospectively held-out set of ten DistilGPT2 lexical fibers, all ten passed a shared-score float64 numerical control and showed a nonzero reward-reference-orthogonal contrast under inverse-length weighting. A later re-audit of the main lexical candidate reduced the alpha-zero numerical discrepancy to 1.33×10^{-16} , resolving the visual ambiguity of earlier float32 residuals while preserving the earlier failed run in the record.

This establishes a precise diagnostic but not a unique mechanism: length-dependent weighting is already well studied. The next experiments therefore move to actual trainer implementations and verifier-defined equivalence.

6.2 Actual Trainer, whole-model transfer, and normalized effects

In the frozen actual GRPOTrainer, five fresh seeds used identical naturally sampled completion tensors in GRPO and a scale-aligned Dr.GRPO reference. Same-arm repeated runs produced bitwise-identical captured gradients under the frozen dropout-off setting. The measured contrast persisted when all 81,912,576 DistilGPT2 parameters were trainable and also in the complement excluding the original 1,536-parameter final-LayerNorm diagnostic subspace.

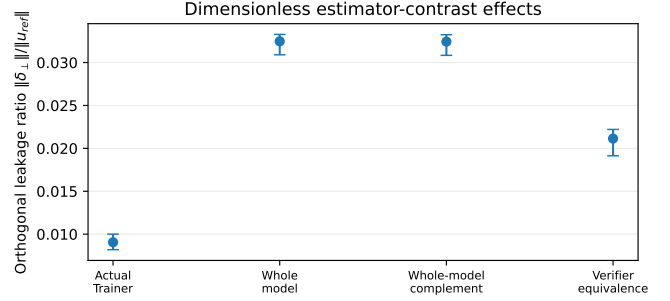


Figure 2: Dimensionless orthogonal leakage ratios in frozen paired experiments. Error bars are 95% bootstrap intervals over independent seeds.

Setting	n	mean $R_{\perp}$	95% seed CI
Actual Trainer	5	0.00906	[0.00819, 0.01000]
Whole model	3	0.03248	[0.03090, 0.03328]
Whole complement	3	0.03244	[0.03084, 0.03324]
Verifier class	3	0.02114	[0.01913, 0.02220]

Table 2: Primary dimensionless effect size. “Whole complement” excludes the original final-LayerNorm diagnostic subspace.

Figure 2 and Table 2 report dimensionless orthogonal leakage ratios. The seed-bootstrap intervals are narrow within each controlled experiment, while the parameter-space scale differs substantially across settings.

6.3 Live current-policy resampling

A 32-step whole-model experiment used 512 fresh current-policy rollouts per step, group size eight, SGD learning rate 10^{-4} , and three fresh seeds. The mean comparative within-fiber log-odds $\Delta F_t = F_{\text{GRPO},t} - F_{\text{Dr},t}$ increased from 0.0376 at step 1 to 0.1677 at step 32 (Figure 3). All final replicates were positive, but trajectories were not uniformly monotone and one seed changed sign at an intermediate checkpoint. Final decoded-objective gaps had mixed signs. We therefore interpret the result as representation-level separation, not evidence of systematic reward harm.

6.4 Verifier equivalence beyond identical strings

For prompt 2+2=, a numeric verifier accepted both "4" (one token) and "+4" (two tokens). Across three paired actual-trainer runs with 16,384 natural rollouts per seed, all runs were valid and produced positive verifier-class fiber projections. The mean orthogonal leakage ratio was 0.02114 (Table 2). This moves the diagnostic beyond exact decoded-string identity while remaining a deliberately simple verifier.

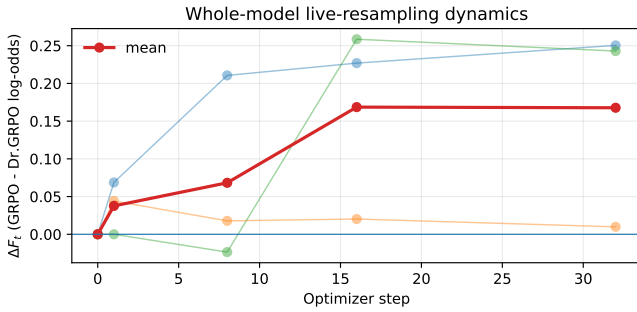


Figure 3: Whole-model live-resampling dynamics. Thin lines show the three replicates; the thick line is their mean. The numerical x-axis uses the actual optimizer steps 0, 1, 8, 16, 32.

6.5 Equal-length clipping as a non-length control

The strongest falsification of a length-only explanation uses prompt Q: 2+2\nA: and two verifier-equivalent one-token representatives: "4" (token 19) and " 4" (token 604). Both have exactly the same token count, verifier reward, and group-relative advantage. A fixed batch is reused after policy lag and standard sequence-level clipping ($\epsilon = 0.2$) is compared with an effectively unclipped arm.

In the prospectively frozen final-LayerNorm phase diagram, selective clipping first appeared at learning rate 0.2: the two ratios were 1.3488 and 0.8181, and the clipped-minus-unclipped second update had relative norm 1.2884 and cosine -0.7225 with the equal-length representation-contrast direction. The prospectively specified whole-model confirmation also separated the two representatives, but with extreme ratios; we therefore treat it only as a causal stress test.

A post-A1 whole-model robustness phase tested a lower-learning-rate grid without changing the original A1 decision rule. Selective clipping occurred at LR 3×10^{-5} on step 3 (ratios 1.3035 versus 1.0151) and at LR 10^{-4} on step 2 (1.6642 versus 1.0094). A LoRA configuration with 405,504 trainable parameters also showed selective clipping at LR 10^{-4} on step 3 (1.2491 versus 0.9728), with a nonzero clipped/unclipped trainable-state difference. Other LoRA learning rates did not yield a nonzero final state difference and are retained as limitations rather than omitted.

Figure 4 summarizes selected equal-length witnesses. Importantly, symmetry-inconsistent PPO clipping is not itself new: Hakoda et al. [2026] derive a coupled symmetric ratio for a symmetric MDP. Here the experiment serves as a controlled demonstration that our quotient-conditioned diagnostic is not reducible to response-length normalization.

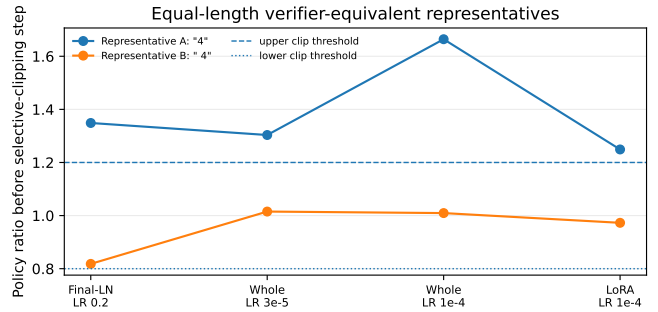


Figure 4: Equal-length verifier-equivalent representatives can occupy different policy-ratio/clipping states. The dashed line is the positive-advantage upper clipping threshold. Settings are selected from prospectively frozen or post-A1 robustness grids and are not pooled as a common effect-size estimate.

Setting	step	ratio A	ratio B
Final-LN, LR 0.2	2	1.3488	0.8181
Whole, LR 3×10^{-5}	3	1.3035	1.0151
Whole, LR 10^{-4}	2	1.6642	1.0094
LoRA, LR 10^{-4}	3	1.2491	0.9728

Table 3: Selected equal-length clipping witnesses. Both representatives are one token and verifier-equivalent.

6.6 Negative ecological pilot

To probe natural prevalence in a less controlled task, we sampled Qwen2.5-0.5B on 80 GSM8K prompts with 16 completions each. The prospectively fixed qualification rule required at least two distinct correct final numeric literals, each observed at least twice. Zero prompts qualified. We do not lower the criterion post hoc. This negative pilot limits any claim about ecological prevalence in base-model reasoning workloads; it does not negate the controlled existence results.

7 Discussion

7.1 What the framework contributes

The combined evidence supports a distinction between *objective equivalence* and *estimator equivalence*. The estimator homomorphism gap in Eq. (11) asks whether the representative-conditioned coefficient $h(y)$ covaries with within-class score directions. Length normalization is one concrete route to such covariance, but not the definition of the phenomenon. Random denominators, policy lag, and clipping can enter the same conditional object.

To our knowledge, the prior work found in our documented search does not combine (i) a representative-conditioned effective-coefficient decomposition, (ii) verifier-defined LLM trajectory quotients, and (iii) direct measurements of the resulting within-class update in a current GRPOTrainer implementation. This is a

bounded literature claim, not proof of global firstness. In particular, the paper does *not* claim novelty for length bias, non-canonical tokenization, MDP homomorphisms, PPO clipping, or clipping-induced symmetry breaking [Ding et al., 2026, Zheng et al., 2025a, Panangaden et al., 2024, Hakoda et al., 2026].

7.2 Interpretation

A nonzero compatibility gap does not imply that the estimator is globally “wrong.” It may have the desired quotient-level direction while also exerting an implicit preference among representatives. That pressure could be harmful, harmless, or useful as an inductive bias. Our results establish and measure the within-class component; they do not establish downstream task-quality harm.

The compatibility perspective may nevertheless be useful for auditing training systems. A trainer can be checked relative to a declared verifier/environment quotient rather than by a mechanism-specific heuristic. This reframes several familiar implementation choices as instances of one conditional invariance question.

8 Limitations

The strongest experiments use relatively small language models and narrow equivalence classes. The original whole-model A1 confirmation uses a high learning rate and produces extreme ratios, although the post-A1 whole-model phase finds selective clipping at substantially lower learning rates. The LoRA transfer is limited: only one tested configuration produced both selective clipping and a nonzero final clipped/unclipped state difference. The GSM8K/Qwen2.5-0.5B ecological pilot was negative under its frozen criterion. Thus prevalence in larger, instruction-tuned reasoning systems remains open.

The number of independent training seeds is modest. Seed-bootstrap intervals summarize variability within the frozen experiments but should not be read as high-powered inferential guarantees. Finally, Theorem 1 is written for scalar per-sample coefficients. Token-local clipping naturally leads to a token-indexed extension and should not be silently treated as covered by the same statement.

9 Conclusion

A reward/environment quotient does not automatically propagate through a stochastic training estimator. For exchangeable score estimators, the representative-conditioned effective coefficient yields an exact quotient/fiber decomposition and a measurable compatibility gap. Controlled tokenization fibers, verifier-defined classes, actual-trainer runs, whole-model dynamics, and equal-length clipping controls show how multiple known estimator mechanisms can produce within-class training

pressure in LLM RLVR. The result is a diagnostic framework rather than a new claim about any single mechanism, and it motivates treating estimator homomorphism compatibility as a separate property of RLVR training systems.

AI-Assisted Research Tools

OpenAI/ChatGPT-family systems assisted literature discovery, code drafting and refactoring, experiment orchestration, numerical cross-checking, and manuscript drafting. The author selected the research questions and claim boundaries, reviewed prospectively frozen protocol boundaries, inspected experiment artifacts and reported results, checked source-level claims used in the manuscript, and approved the final manuscript. AI systems are not authors or sources of experimental evidence.

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A Proof Details and Exact Audit

A.1 Local necessity under tangent-surjectivity

For a quotient class z , define the conditional score-tangent operator

$$T_z(v)(y) = v^\top s_F(y | z). \quad (16)$$

Suppose its image contains the centered coefficient

$$\tilde{h}(y) = h_\theta(y) - \bar{h}_\theta(z). \quad (17)$$

Then there exists a tangent v with $v^\top s_F(y | z) = \tilde{h}(y)$. Hence

$$v^\top \Gamma_\theta(z) = \mathbb{E}[h(Y)\tilde{h}(Y) | z] \quad (18)$$

$$= \mathbb{E}[\tilde{h}(Y)^2 | z] \quad (19)$$

$$= \text{Var}(h(Y) | z). \quad (20)$$

If h is nonconstant within the class, the variance is positive. Therefore class-constancy is locally necessary whenever the within-class score tangent space can represent the centered effective coefficient. It is not claimed necessary for arbitrary underparameterized policies.

A.2 Group-mean equal-weight control

If rewards depend only on $Q(Y)$, $A_i = R_i - \bar{R}$, and every sample has the same fixed scalar coefficient c , then conditional on $Y_i = y$ the expected group advantage depends on y only through $Q(y)$. Thus $h(y) = c a(Q(y))$ for a quotient-level scalar a , and the estimator is class-constant within each quotient class.

A.3 Random denominator

A group-global denominator can still be representative dependent in conditional expectation. With

$$H_i = \frac{A_i}{S_B}, \quad S_B = \sum_j T(Y_j), \quad (21)$$

conditioning on $Y_i = y$ fixes one contribution $T(y)$ to the random denominator. Therefore $\mathbb{E}[A_i/S_B | Y_i = y]$ may vary with the representative even though the same denominator is shared by all samples in a realized batch.

A.4 Exact enumeration audit

An independent finite audit randomized three two-representative quotient classes and exact-enumerated all batches for $B \in \{2, 3, 4\}$ across 750 trials. The maximum decomposition error was 5.65×10^{-15} ; the maximum class-constant control error was 1.28×10^{-14} ; and all two-representative local nonconstant-coefficient directional witnesses passed. The tolerance was 10^{-12} .

B Implementation and Measurement Details

Actual-trainer direct gate. DistilGPT2 used prompt `Answer:` with reward one only for terminal decoded `"\n\n"`. The two rewarded paths were token 628 (length one) and tokens 198,198 (length two). Five fresh seeds used group size eight, 1,024 natural rollouts, microbatch 64, gradient accumulation 16, SGD LR 0.01, no reward standardization, beta zero, and dropout disabled. Scientific vectors were captured immediately before the optimizer step. Every same-arm repeat was bitwise identical.

Whole-model dynamics. All 81,912,576 parameters were trainable. Three fresh seeds used group size eight, 512 fresh current-policy rollouts per step, 32 optimizer steps, SGD LR 10^{-4} , microbatch 64, gradient accumulation eight, no momentum, no gradient clipping, no reward standardization, beta zero, and dropout disabled. Informative rollout calls remained non-saturated for essentially the full trajectory.

Verifier-equivalence gate. Prompt `2+2=` used a verifier that accepted exactly `"4"` and `"+4"`. Three fresh seeds each used 16,384 natural rollouts and group size eight. Counts of the two rewarded strings were respectively (413,14), (420,11), and (445,8). No other path received reward one.

Equal-length A1. Probability-only candidate selection occurred before gradient inspection. The selected representatives were `"4"` (token 19, natural probability approximately 0.00106) and `" 4"` (token 604, approximately 0.01805), both one token and both accepted by `completion.strip()=="4"`. Every group contained both positive representatives and six fixed one-token reward-zero distractors, so both positives had equal group-relative advantages and all completions had equal length.

The first A1 diagnostic artifact used a diagnostic old-policy log probability captured before Trainer dropout disabling. That artifact was declared implementation-invalid and retained. The repaired run bound diagnostics to the exact old-policy values returned after dropout disabling; no scientific threshold, candidate, or learning-rate grid was changed.

C Falsification and Negative Results

We retain the following failures and invalid runs rather than silently dropping them:

- **Cross-fiber v0.1 (failed validity gate):** separate float32 autograd equality residuals invalidated 6/10 controls. The run remains failed; a new unseen v0.2 holdout was used instead of rescoring.

- **BNPO closed-loop directional prediction (failed):** the final BNPO–Dr.GRPO mean fiber gap was -0.004084 , rejecting a broad long-run directional claim.
- **Actual Trainer v0.1 (invalid):** active dropout confounded the implementation comparison.
- **Actual Trainer v0.2 (invalid):** a prospectively frozen separate numerical-fidelity oracle was too strict for the implementation-level comparison; it was not rescored.
- **Natural lexical-suite discovery (negative):** no additional DistilGPT2 lexical surfaces passed the frozen high-mass threshold after exclusions.
- **Whole-model dynamics v0.1 (limited):** LR 0.01 rapidly saturated the decoded reward, so it is not used as evidence for 32 informative accumulation steps.
- **A1 first diagnostic artifact (invalid):** old-policy diagnostics were captured before Trainer dropout disabling; repaired measurement preserved the scientific protocol.
- **Qwen2.5-0.5B/GSM8K ecological pilot (negative):** 0/80 prompts met the frozen criterion of two distinct correct numeric literals each observed at least twice among 16 samples.

D Prior-Art Boundary

Broad response-length and aggregation effects are occupied by Dr.GRPO, ΔL Normalization, Balanced Aggregation, and the impossibility analysis of Ding et al. [2026]. Non-canonical tokenization is occupied by work such as Zheng et al. [2025a]. MDP homomorphism and homomorphic policy-gradient theory are established [Rezaei-Shoshtari et al., 2022, Panangaden et al., 2024]. PPO ratio clipping is classical [Schulman et al., 2017], sequence-level clipping in LLM RL appears in GSPO [Zheng et al., 2025b], and Hakoda et al. [2026] directly analyze symmetry consistency of PPO with coupled versus per-side ratios.

Accordingly, our novelty claim is deliberately narrow. We do not claim first discovery of clipping-induced symmetry breaking, length bias, non-canonical tokenization, or policy-gradient homomorphisms. The retained contribution is the quotient-conditioned effective-coefficient diagnostic, its exact decomposition, and its operationalization on verifier-defined LLM trajectory classes in actual trainer executions.

E Reproducibility Record

The actual-trainer experiments used a vendored TRL source frozen at commit

`a98fa6a4428f9aae58dfb26d729d7437f662f27a`

with `grpо_trainer.py` SHA-256

`e9f5ad165c620c8da1eceede84b00e405`
`c76691d9af363681e8f7cbbfae8a709.`

The paper-level evidence set is maintained with SHA-256 provenance and includes positive, negative, and invalid runs. A public reproduction package is intended to accompany the public release; private patent/product notes and unrelated local-workspace material are excluded from the scientific package.

The models referenced in the paper are DistilGPT2 [Hugging Face, 2026a] and Qwen2.5-0.5B [Yang et al., 2024]. The current TRL software surface is documented by Hugging Face [Hugging Face, 2026b].